

Issues in the Adoption of the Scaled Agile Framework

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Abstract

Agile methods were originally introduced for small sized, co-located teams. Their successful products immediately brought up the issue of adapting the methods also for large and distributed organizations engaged in projects to build major, complex products. Currently the most popular multi-teams agile method is the Scaled Agile Framework (SAFe) which, however, is subject to criticism: it appears to be quite demanding and expensive in terms of human resource and project management practices. Moreover, SAFe allegedly goes against some of the principles of agility.

This research attempts to gather a deeper understanding of the matter first reviewing and analysing the studies published on this topic via a multivocal literature review and then with an extended empirical investigation on the matters that appear most controversial via the direct analysis of the work of 25 respondents from 17 different companies located in eight countries. Thus, the originality of this research is in the systemic assessment of the “level of flexibility” of SAFe, highlighting the challenges of adopting this framework as it relates to decision making, structure, and the technical and managerial competencies of the company.

The results show that SAFe can be an effective and adequate approach if the company is ready to invest a significant effort and resources into it both in the form of providing time for SAFe to be properly absorbed and specific training for individuals.

CCS Concepts

• **Software and its engineering** → **Agile software development**; *Collaboration in software development*.

Keywords

Empirical software engineering, distributed agile, surveys

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1 Introduction

Agile software development methods started emerging in mid-1990s as an alternative to traditional, “plan-driven” approaches [1]. Agile methods were originally conceived for small and co-located teams, where a shared vision and accountability are easier to achieve [64]. However, the diffusion of agile practices lead to the attempts to adapt them to diverse, large and distributed organizations, with the consequent issues on how to adapt them [15, 30]. Moreover, from the very beginning the problems about integrating agile teams within large organizations were reported. Even the prototypical agile project, the C3 project done under the leadership of the groundbreaking innovator Kent Beck, and featuring a team of very expert software engineers, ended with the team being fired by Chrysler [25].

Scaling of agile methods to multiple teams requires the solution to a variety of challenges, including which roles change or have to be introduced, how to synchronize and coordinate the work of different teams, and how to handle the overall communication process [60]. Several scaling frameworks have emerged during the last few years, such as SAFe, LeSS, DAD [30]. The most popular one, according to the State of Agile annual report, is SAFe [18].

1.1 Rationale for the research

As companies attempt transformation towards SAFe, a series of problems emerged that have been discussed in the literature, the most relevant being [19]:

- SAFe is too complex and hard to understand [22], and
- it creates a top-down organizational control that leads to a waterfall-like structure, which ends up being not agile and with limited flexibility [39].

Our study aims to investigate such criticisms, the consequences of SAFe application, and the overall factors that affect a successful transformation towards SAFe. Specifically, the research questions presented in this paper are:

RQ₁: What are the most crucial success factors for adopting SAFe?

RQ₂: Could SAFe hinder agility, and, if so, why and how?

RQ₃: What are the typical effects of SAFe on large organizations?

Our investigation takes a double-step approach. First, we perform a multivocal literature review to systematically collect the current wisdom about SAFe. Then, on such basis, we perform a focused

observational analysis on it reviewing the work of 25 SAFe practitioners from 17 companies from 8 countries. We employ simple statistical techniques, as the size of the sample makes it unsuitable for inferences.

The originality and the importance of our work lies on being the first one assessing systematically, both through a multivocal literature review and empirically, the “level of agility” of SAFe across multiple companies located in multiple countries, which is reported as being a very compelling issue for developers [23, 37] (see Section 2). Moreover, our findings present a new perspective on the matter focusing the problem of adopting SAFe on the overall decision making, on the structure, and on the technical and managerial competence of the company.

Indeed, we argue that the third major contribution of our work is that, rather than presenting conclusive results, it opens the discussion to analysing the problem under a different perspective, in the hope of having followup discussions, other empirical studies, and replicas of this one, thus generating a substantial corpus of knowledge on this very important matter.

This paper is organized as follows. In Section 1.2 the brief overview of SAFe framework is given. Section 2 performs the mentioned multivocal literature review and Section 3 summarises the current state of the art of the technical and scientific investigations. Section 4 details the structure of the empirical investigation, whereas its results are presented in Section 5. Section 6 provides consolidated results for both areas of the study with respect to the three key research questions. Section 7 discusses the internal and external validity of the results. Section 8 draws some conclusion and outlines the future research.

1.2 Background on SAFe

SAFe proposes an approach that targets large organizations: it details roles, responsibilities, structures, and activities using a combination of agile and lean development, DevOps, and system thinking [58]. Note that in this paper, a large company refers to an organization with 250 or more employees.

At the top of its approach SAFe defines a set of core values [58]:

- Alignment (with the business of the company)
- Built-in quality (of the products)
- Transparency (of the development process)
- Program execution (for building manageable value streams)

To handle the integration of agility within large and possibly distributed organizations, SAFe organizes its structure in four levels: team, program, large solution, and portfolio (Figure 1).

At the “team level,” teams — similar to Scrum teams including a Product Owner, a Scrum Master, and a few developers — operate fully with an agile approach, including the delivery of software increments at each short iteration. In particular, each team is allowed to freely choose for its internal process any of Scrum, Kanban, or Scrumban. The “program level” integrates different teams up to a total of 150 people and releases their work incrementally in 8-12 weeks long sprints with the metaphor of the *train* (the Agile Release Trains or ARTs). Some key roles added are: the Product Management, who is responsible for the product backlog at this level, and the Release Train Engineer, who is responsible for coaching the teams and their Scrum Masters. Large projects not fitting the program level are then coordinated in the “large solution” level,

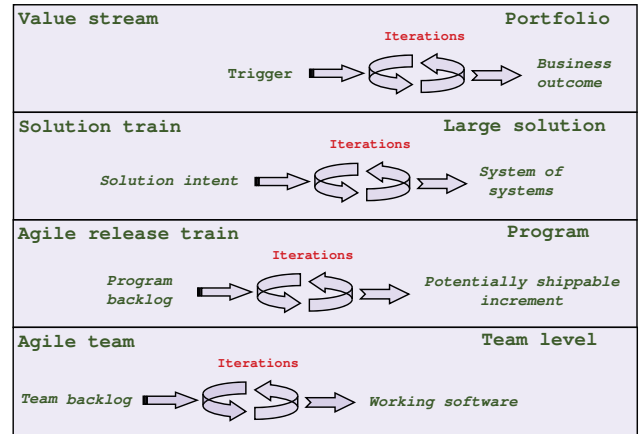


Figure 1: Structure of SAFe, taken with modification from [44]

where there are specialized roles handling the extra complexity and size of such projects, e.g., Solution Architect, Solution Management, and Solution Train Engineer. Eventually, the “portfolio level” defines the strategic priorities and directions of the ARTs and allocate their budgets. Notice that not all levels are needed. SAFe can be structured in different configurations of levels depending on the size and the structure of the target organization [58].

2 Structure of the multivocal literature review

Researchers and practitioners of agile methods produce a large amount of gray literature, which is particularly valuable despite the fact that it might be disregarded in more “structured” research forums [28, 56]. To gather a comprehensive summary of the current understanding of the effect of the practices of SAFe we have performed a multivocal literature review (MLR) [28]. For this reason, in our MLR we have also included studies coming from gray literature, following the same approach described in [57].

In this MLR we have followed the protocol defined by Kitchenham [40] and Garousi [28] (Figure 2).



Figure 2: Steps of the MLR

2.1 Definition of search string

To define the search string at the core of our MLR, we have performed an iterative search process that started from a limited search in Google Scholar [28]. Furthermore, the overall web was searched with the Google search engine for incorporating gray literature using a reduced search string. The search strings are presented in Table 1.

2.2 Selection of sources

The search was defined and performed on the major databases of technological and scientific literature: IEEE Xplore, ACM Digital Library, Google Scholar. In addition, the following journals and websites were hand-searched for the relevant gray literature: Medium (medium.com), Agile Alliance (www.agilealliance.org), SAFe official website (www.scaledagileframework.com), AgileCraft

Table 1: Search strings

Type	Search string
White literature	(SAFe OR "scaled agile framework") AND ("large agile" OR "large-scale agile" OR "scaling agile") AND ("success factor*" OR "success" OR fail OR "fail factor*")
	SAFe OR "scaled agile framework" AND ("large agile" OR "large-scale agile" OR "scaling agile") AND (advantages OR disadvantages OR benefits OR drawbacks)
	(SAFe OR "scaled agile framework") AND ("large agile" OR "large-scale agile" OR "scaling agile") AND ("agility")
	Scaled agile framework,
	Agile industrial complex,
Gray literature	large organization agile transformation,
	scaled agile framework fail,
	scaled agile framework success,
	scaled agile framework effects on agility

(blog.agilecraft.com), Agile annual reports (www.stateofagile.com), Mike Cohn's blog (www.mountaingoaftsoftware.com/blog), Ken Schwabber blog (www.scrum.org/user/84), Martin Fowler blog (martinfowler.com), David J Andersen blog (djaa.com/blog)

2.3 Inclusion and exclusion criteria

After searching and retrieving the set of potentially interesting papers, we have applied the following inclusion and exclusion criteria on such set [48]:

- **Inclusion criteria:**

- *Type of intervention:* Articles that focuses on consequences and impact on organization of applying SAFe. Articles that focuses on fail/success factors of applying agile scaling methods described above.
- *Type of outcome:* A success and fail stories of applying SAFe. Additionally empirical outcomes about organizations transformation during applying this method and impacts on agility.
- *Topic/scope:* SAFe in large organizations
- *Date:* Published after 2009
- *Language:* English
- *Type of publication:* Original researches, review researches, case studies, blogposts, whitepapers, reports
- *Settings:* Only large organizations (organization with 50 or more people or at least 5 teams)

- **Exclusion criteria:**

- Failure to meet any one of the above inclusion criteria will result in exclusion from the review pages)
- Papers and reports will be excluded where only the abstract, but not the full text, is available.

2.4 Quality criteria

The criteria that were used to assess the quality of each "white" publication were taken from [21] and they are:

- (1) Is the paper based on research?
- (2) Is there a clear statement of the aims of the research?
- (3) Is there an adequate description of the context in which the research was carried out?
- (4) Was the research design appropriate to address the aims of the research?

Table 2: White literature quality scores

Ref.	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Tot.
[50]	1	1	1	1	1	0.5	1	0.5	1	1	9
[56]	1	1	1	1	1	1	0.5	1	1	1	9.5
[12]	1	1	1	0.5	1	1	1	0.5	1	1	9
[55]	1	1	1	1	1	1	1	1	0.5	1	9.5
[49]	1	1	1	0.5	1	0.5	1	0.5	1	0.5	8
[3]	1	1	1	0.5	1	0.5	1	0.5	1	0.5	8
[50]	1	1	1	1	1	1	1	1	1	1	10
[51]	1	1	1	0.5	1	0.5	1	0.5	1	1	8.5
[53]	1	1	1	1	1	1	1	1	1	1	10
[43]	1	1	0.5	1	1	0.5	0.5	1	1	1	8.5
[26]	1	1	1	1	1	1	0.5	1	1	1	9.5

Table 3: Gray literature quality scores

Ref.	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Tot.
[33]	1	1	0.5	1	0	1	0	0.5	5
[37]	1	0.5	0	1	0	1	0.5	0.5	4.5
[38]	1	1	0.5	1	1	0	1	0.5	6
[2]	1	1	1	0.5	0.5	0	0.5	0.5	5.5
[23]	1	1	0.5	1	0.5	1	0	0.5	5.5
[31]	1	1	0.5	0.5	0.5	0.5	0.5	0.5	5
[58]	1	1	1	1	1	1	1	1	8
[18]	1	1	1	1	0.5	0.5	0.5	0.5	6

- (5) Was the recruitment strategy appropriate to the aims of the research?
- (6) Was the data collected in a way that addressed the research issue?
- (7) Was the data analysis sufficiently rigorous?
- (8) Has the relationship between researcher and participants been adequately considered?
- (9) Is there a clear statement of findings?
- (10) Is the study of value for research or practice?

The criteria for gray literature were derived from [29]. They are:

- (1) Authority of the producer
- (2) Methodology (Does the source have a stated methodology?)
- (3) Objectivity (Does the work seem to be balanced in presentation?)
- (4) Date (Does the item have a clearly stated date?)
- (5) Position w.r.t. related sources
- (6) Novelty
- (7) Impact
- (8) Outlet type (Books, annual reports, wiki articles, blogs, etc.)

The answers to each question were one of the following values: 1 (meaning 'Yes'), 0 ('No') or 0.5 ('Maybe'). All calculated quality scores are shown in Table 2 and Table 3.

2.5 Data extraction and synthesis

These criteria have then applied in three phases leading to a final selection of 19 papers (Table 4):

Phase 1: All articles were processed based on title and abstract.

Phase 2: The result was refined considering introduction and conclusion.

Phase 3: A final selection was performed based on full text analysis.

Table 4: Filtering results

Source	Phase 1	Phase 2	Phase 3
Google	27	27	7
IEEEExplore	6	4	1
ACM DL	6	4	2
Google Scholar	133	45	8
SAFE official	3	2	1
Total	178	74	19

The final outcome of the search process is presented in Table 5.

Table 5: Results from the MLR

Ref.	Approach	Description	Type	#
[33]	blogpost	project manager shares her experience of transformation towards SAFe	gray	1
[23]	blogpost	product leader's view on philosophical issues of SAFe	gray	-
[58]	whitepaper	overview of SAFe	gray	-
[31]	blogpost	discussion of problems and possible solutions of SAFe	gray	1
[18]	survey	survey of agile practitioners all over the world	gray	1
[56]	multivocal literature review	analysis of transformation towards SAFe based on the cases from SAFe official website	white	42
[49]	case study	pros and cons of SAFe and how it differs from waterfall	white	1
[50]	qualitative case study	challenges and success factors of transformation towards SAFe in two different business lines inside one company	white	2
[12]	case study	analysis of success factors and effects of transformation towards SAFe in two different companies in Finland	white	2
[55]	action case study	experience of transformation towards SAFe — challenges and advises for future users	white	1
[26]	survey	SAFe analysis with focus on test and quality	white	17
[3]	case study	factors behind successful transformation towards SAFe	white	2
[53]	case study	effects and challenges of transformation towards SAFe	white	1
[51]	journal paper	view on SAFe drawbacks	gray	-
[43]	survey	benefits and problems of transformation towards SAFe in Finland	white	111
[2]	blogpost	agile coach's view on SAFe and SAFe's official course	gray	-
[37]	blogpost	Ron Jeffries's analysis of SAFe issues with improvements suggestion	gray	-
[38]	blogpost	Ron Jeffries's analysis of SAFe SPC course	gray	-

Then, we have analysed the results following the guidelines given in [40]:

- the documents were clustered based on the research question to which they were related; each study could be related to several research questions;
- the data for each research question and the papers related to the question were differently tabulated;
- the data were transformed to a common format and grouped, to find the links between the studies;
- the data were translated to identify the common areas among the presented studies.

3 Results of the multivocal literature review

3.1 RQ₁: Success factors of SAFe

The analysis of the literature has identified 6 different success factors for the implementation of SAFe present in at least 3 cases (see Figure 3).

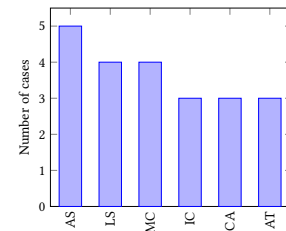


Figure 3: Success factors in SAFe: Appropriate Staffing, Leadership Support, Measuring Changes, Involvement and Communication of all parties, Change Agents, Agile Training for employees.

Indeed, as observed in [12, 55], most of them are related to the overall infrastructure provided by organizations to their employees for succeeding in SAFe, such as appropriate staffing, the direct and positive motivation of the managers, the presence of systematic ways to measure and track SAFe (as also recommended directly inside SAFe) [35, 42, 58].

Also the inclusion of motivated team members, who already have or are gaining competence in agile and in SAFe, appeared of paramount importance: SAFe — like practically all lean processes in knowledge intensive organizations — requires people who become actively involved in improving and transforming constantly their behavior [2, 3, 49] and it is very important that the organization establishes suitable coaching for those employees who suffer more difficulty [55].

3.2 RQ₂: Could SAFe hinder agility?

The relationship between SAFe and agility overall is indeed controversial; there are aspects in SAFe that appear to conflict with agility and in the considered literature we have found five of such issues mentioned more than once (see Figure 4).

Specifically, some of the practitioners in the gray literature have found that the use of SAFe leads to an inability to improve [23, 37], because the structured process proposed by SAFe reduces the employees' ability to continuously learn and adapt processes based on their experiences and because the value of (*alignment*) is incorrectly implemented in SAFe leading to the inability to tailor the structure to the specific needs arising in every situation.

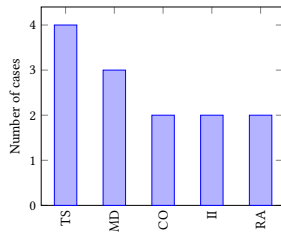


Figure 4: Aspects of SAFe conflicting with Agility: Team Structure, Methodology too Detailed, Complexity and Overhead, Inability to Improve, Requirements for Alignment.

3.3 RQ₃: Effects of SAFe

10 benefits and 7 drawbacks were mentioned multiple times in the target documents. The mentioned benefits and drawbacks are shown in Figure 5 and Figure 6 respectively.

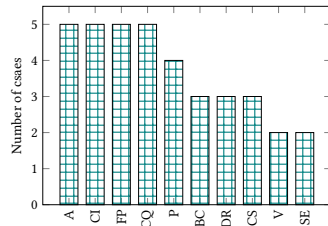


Figure 5: Mentioned positive effects: Alignment, Continuous Improvement, Frequent and Predictable deliveries, improved Code Quality, Productivity, Better Communication, Dependencies Resolution, Clear Structure, Visibility, Satisfaction of Employees.

The benefits seem to be lead by the core values of SAFe and to be mostly linked to the business side of the organization, even if there are also mentions of satisfaction of employees.

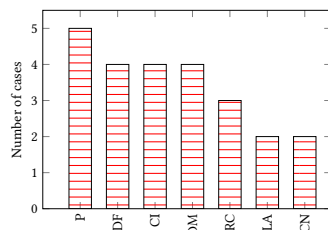


Figure 6: Mentioned drawbacks: Planning, Difficulty of the Framework, challenges for Continuous Improvement, hierarchical Decision Making, Resistance to Change, Lack of Autonomy, Communication with Non agile parts.

The biggest challenges in SAFe appears to be linked to the alleged overall complexity and rigidity of its structure, which in turn is reflected into the opinion shared by some practitioners that SAFe is not agile. This then is also reflected into the resistance to change reported in some of the studies from Finland [43, 53] and in alleged inability to handle fast changes in requirements.

3.4 Summary of the MLR findings

Table 8 summarised the results of the multivocal literature review. Overall, we can see that the current work tend to confirm that SAFe requires a substantial commitment of the company senior

management, and competent and motivated workforce. This has also an effect on agility; indeed, the mindset of some managers tend to be skewed toward organization-wide structures and centralized control mechanisms that inherently conflict with agility, as observed in [36]. SAFe appears to promote aspects that appeal to traditional managers and poses issues related to some of the perceived core-values of agility.

The current understanding of SAFe raises the concern that different assessments appear to be originated on the background on the person making such assessment, especially in the case of gray literature, where, in simple terms, we can see two opposite positions:

- those coming from originators or enthusiasts of agile methods, who tend to emphasize the problems connected with the alleged difficulty, and flexibility and adaptability to change;
- those with a more “standard” managerial background, who tend to appreciate the structured and pervasive approach of SAFe.

4 Structure of the empirical investigation

From the analysis of the literature it appears that there is a need to strengthen the current and mostly anecdotal and somehow biased understanding of the research questions with empirical evidence. For this reason we moved ahead with the research using an open questionnaire as our key research instrument, which is known to be the most suitable approach for exploratory research [67]. Designing an effective questionnaire for an interview about interaction and communication is not easy, since the way questions are defined, organized, and laid down may significantly influence the answers [16]. Therefore, we followed the rules defined in [9, 27, 54, 63] to minimize the biases in the responses, including redundancy and replication.

Participants for the survey were hunted using LinkedIn profiles’ search, where the candidates were searched by specific SAFe positions or by belonging to companies that use SAFe [47].

The interviews were conducted online and recorded with permission of the interviewee and then the result was processed and sent back for verification.

The survey was created first defining an overall GQM of the research goals [4], and then drafting two prototypes of questions [34]. First, we defined the goal of the interview: what information we wanted to retrieve. Then, we drafted the questions and conducted two rounds of pilot interviews. We revised the wordings of the questions based on the results of the pilot conversations and created the final version of the questionnaire, proceeding with significant care and following the best practices in the field given the issues associated also with online interviews [6, 46].

The script of the interview questionnaire is given in Table 6. The structure followed the recommended standards [7, 11]. The first part consists of background questions, which were added to retrieve some contextual information about the role in transformation and experience with agile. Additionally, these questions helped us find information about the current state of transformation: exploring, processing or mastering. The next part consists of Likert scale survey. Each question summarizes the reasons why SAFe breaks agile values — based on the findings of MLR. The scale reveals

Table 6: Interview questionnaire

	Questions
Background	1. Name
	2. Company name
	3. What is your current role?
	4. How long do you work in organization? In this team?
	5. Have you been in team before transformation?
	6. How many year experience in software development?
	7. What is your previous experience with agile?
	8. How large is your team?
	9. What is state of SAFe transformation?
RQ1	10. Name the factors that helped in your work during agile transformation
	11. Name the factors that interfered your work during agile transformation
	12. How these factors helped/interfered you?
RQ2	13. Do you agree with statement: SAFe provides comprehensive model of management and workflow, that complicates the whole process
	14. Explain your answer
	15. Do you agree with statement: It concentrated more on methods and processes to implement them with discipline rather than fit them in the context of the organization
	16. Explain your answer
	17. Do you agree with statement: SAFe does not let teams to learn and improve their process exploiting their experience
	18. Explain your answer
	19. Do you agree with statement: Alignment (in SAFe) means "everyone do this" and the direction pushed by management. This leads to inability of teams to tailor structure and "move in better direction"
	20. Explain your answer
RQ3	21. Name the benefits that SAFe brings to project and your work
	22. Name the drawbacks that SAFe brings to project and your work
	23. How this benefits/drawbacks was measured? What metrics was used?

the level of agreement of the interviewee with the statement. The fourth part is a set of open-ended questions.

The population of our research was restricted by employees of companies that implemented or implementing SAFe. Using the four levels of SAFe as a guide, four groups were created and sample clustering was employed on the four groups [45]. We chose interviewees inside clusters randomly to reduce an influence of specific position view on results [59]. The quality of the results was improved with triangulation by data source was used for improving the quality of the results [65]. We tried to access participants from different companies, in different positions and from the team to top management level [52].

The responses were analysed using fundamental statistics on ordinal scales [10] and windowing and coding the answers [16]. The study is mostly observational; as usual in software engineering it is difficult to find subjects willing to join a study like this one,

and this is even more challenging because our focus is the small set of software managers and engineers using or interested in SAFe. Therefore, we will not attempt to draw any statistical inference on our results. Still our claim is that our results are particularly important as they evidence a trend of ideas that could lead to solid conclusions after further analyses and/or replicas.

5 Results of the empirical investigation

5.1 Participants

As mentioned we have distributed our questionnaire on technical social media and we have received 25 responses out of 136 requests, with a response rate of around 18%, which appears quite standard. The participants come from all over the world with a higher incidence from US (9 participants) and Russia (3), which is quite understandable due to the higher presence of industry, then we have UK (3), Australia, Canada, Serbia, Spain, and Switzerland (1 each), and 3 did not specify their location. The participants covered all roles of SAFe but with different levels (Table 7). The participants have an average working experience of about 15 years and they are from about 22 years of age to about 60 with an average value of around 40.

Table 7: Role in SAFe

Role	Number of participants
Release Train Engineer	14
Scrum Master	3
Project manager	2
Agile coach	1
Product owner	1
Release Manager	1
Software developer	1
Solution Train Engineer	1
Technology Program Manager	1

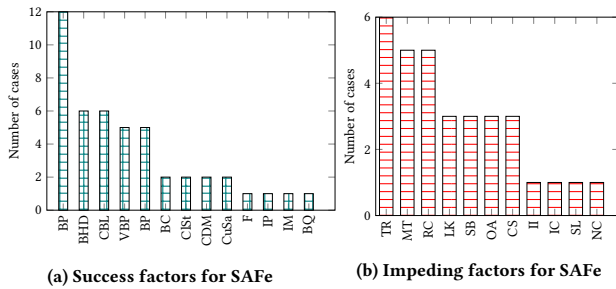
5.2 Results

5.2.1 RQ1: Success factors of SAFe

The participants were asked to list the key success factors of SAFe and its major inhibitors with an open question. The resulting success factors for SAFe are given in Figure 7a.

What we see here is a result that on one side correspond to what we found in the literature (Section 3.1): the support of the leadership with the consequent presence of adequate staffing and competences is present. However, the responses taken together emphasize that the success depends on the establishment of a process that is fully understood by all parties. The leadership of a company has the opportunity of having an agile method that is comprehensively managed as one with the company as a whole, but this requires the trust to the people down the hierarchy, so coaching is essentially, like creation of a shared vision, moving the decisions down the hierarchy of SAFe as much as possible, being transparent in the decisions that are taken, and so on [13, 41].

The impeding factors appear to be absolutely congruent with the success factors and also reflect what was evidenced in the "15th Annual State of Agile Report" [18]. The resulting major obstacles of SAFe are given in Figure 7b.

Figure 7: RQ₁ – success factors and impeding factors of SAFe:

Success factors: Leadership Support in transformation, strong synergy of Business and Development, Coaching, appropriate Culture and Mindset, Competent Staff, Pilot Teams, Product Vision, Well-motivated Choices, moving Responsibilities Down, Communication between Layers, Support from Teammates, Relevant Knowledge, Trust between Teams.

Impeding factors: lack of Time and Resources, wrong Mindset of the Top management, Resistance to Change, Lack of Knowledge within the company, SB: Skepticism of the Business side, conflict with Old Approach, Complex Structure, Incorrect Implementation, Inappropriate Culture, no Support from Leaders, No Coaching.

Again, we see that a comprehensive and fully informed support from the top management and a clear and deep understanding of, and commitment to SAFe in all the part of an organization are a prerequisite for its successful transformation, and lack of any part of them can be detrimental. Well, in general any process and any change in the business of a company require support and understanding, but here it appears that it is even more crucial. The impression that we are receiving is that SAFe is very fragile with respect to the lack of support, commitment, and technical and managerial competence in the company.

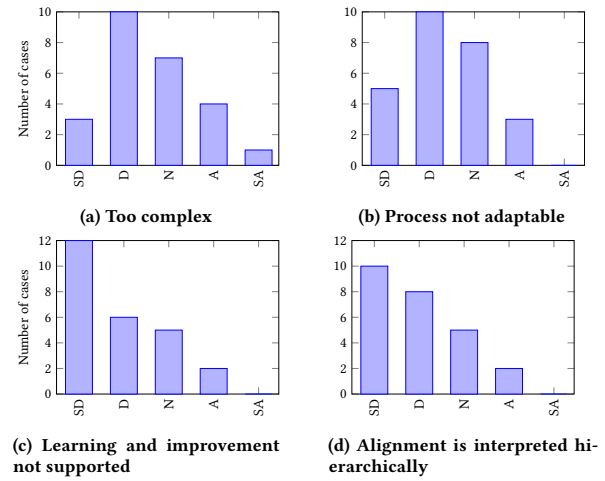
5.2.2 RQ₂: Could SAFe hinder agility?

To target RQ₂ we have asked the level of agreement on a 5 point Likert scale on four possible ways of hindering agility:

- it is complex (see Figure 8a);
- it focuses more on following predefined processes than on adapting processes to the need of the organization (see Figure 8b);
- it does not support learning and improvement (see Figure 8c);
- it interprets alignment in a prescriptive and hierarchical way (see Figure 8d).

The summary of all answers is in Figure 8. Some of the participants did not answer to all the questions, however this analysis is mostly qualitative and without inferential statistics; therefore, the use of censored data does not affect significantly the conclusions for Likert scores on an ordinal scale [5].

The overall results show that in general, considering the median of the answers, the interviewees do not agree on any obstruction of agility. However, going into details, we notice that the complexity of SAFe (see Figure 8a) and its rigidity are perceived in general as more challenging (see Figure 8b), while a continuous learning and improvement (see Figure 8c) and alignments (see Figure 8d) do not appear to pose significant issues.

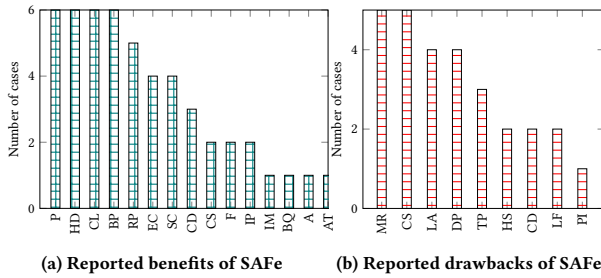
Figure 8: RQ₂ – aspects of SAFe that may hinder agility. 5 points Likert scale: Strongly Disagree, Disagree, Neutral, Agree, Strongly Agree

A possible explanation of this lies again on the comprehensive commitment required by SAFe, of which we have discussed in Section 5.2.1. SAFe appears to require a significant amount of upfront dedication from the whole company. This dedication translates on the fact that initially people might have to go over a feeling of being overwhelmed by its complexity, quite like the writers of the blogposts we mentioned in the analysis of the literature review. After people pass this stage overall, as an organization, then, the approach appears to be effective. This also relates to the fragility of the approach, a lack of such commitment on any side of the organization might trigger and overall failure. Finally, also this analysis is completely compatible with the current state of the art discussed in Section 3.2, where the highest scores related to organizational structure and the complexity of the approach, while the opportunities for improvement and the issues related to alignment appear less remarkable.

5.2.3 RQ₃: Effects of SAFe

Eventually, we have asked the participants to list the benefits and the drawbacks they perceive in SAFe (see Figure 9). Taking into account the comments that we made before, the listed benefits are perfectly compatible both with the benefits we identified in the review of the literature (Section 3.3) and the ones present in the mentioned SAFe whitepaper [58]. Specifically the biggest reported benefits relate to a better management of the company as a whole, that is, overall increased productivity, shared vision, coordination of work (which also means alignment), interaction between the different levels in the company and so on.

Also the most commonly identified drawbacks are fully compatible with what we have read in the literature and can be interpreted in the sense discussed above, that is, that SAFe requires a major commitment at all levels of an organization, time to set up, and it is fragile. In this sense we interpret the need for more resources, the more complex structure, the alleged lack of autonomy, and so on.



(a) Reported benefits of SAFe (b) Reported drawbacks of SAFe
Figure 9: RQ3 – reported benefits and drawbacks of SAFe:

Benefits: better Productivity, better Handling of Dependencies, Coordination between Levels, vision of the Big Picture, Reliable Predictions, more Effective Communication, better Structure of Code, Coordinated Decisions, Customer Satisfaction, Flexibility, Improved Prioritization, Increased Motivation, Better Quality, Alignment, Autonomous Teams.

Drawbacks: requires More Resources, Complex Structure, Lack of Autonomy, Decreased Productivity, Top-down Planning, Higher level of Stress, Customer Dissatisfaction, Lack of Flexibility, Prioritization of Issues.

6 Overall findings

Considering the results of our empirical investigation we can provide some comprehensive answers to our research questions.

As for **RQ₁** about the success factors in SAFe it is evident that the first and foremost reason for success of SAFe lies in a comprehensive and informed support from the organization at all its level. It could be objected that this is true for any methodology, but in this case, this is even more important. We can see that all the advantages and the problems related to SAFe really lie into this support. Indeed, support means also being able to invest upfront in education and to provide time for the methodology to be learnt properly, as additional success factors relate to this.

As for **RQ₂** about whether SAFe hinders agility, we can provide a twofold answer. First, overall, there is no agreement on whether SAFe really hinders agility, as the data we have collected does not provide even in one case an average answer supporting this it, and anyway, in the literature we can find this claim referring to situations where not enough commitments was provided to SAFe. However, we do see that SAFe requires a significant upfront commitment, including time and resources, and that it is quite fragile. Under these perspectives, there is a conflict with the paramount ideas that agility should require lower upfront investments and being a quite robust methodology. However, we have to say that throughout our studies we have found that, in contrast to the belief of many, agile environments do require a comprehensive support from the organization as a whole and require higher level of commitment from participants, and in these senses are more challenging to implement, require upfront resources, and are more fragile [14]. Under this perspective, we can say that SAFe does not hinder agility, but actually exposes the real nature of it.

Eventually, with reference to **RQ₃** about the effect of SAFe, we do see here completely coherent statements with what we have discovered so far, which is also completely compatible with the earlier findings. Substantially, SAFe is perceived to produce significant positive effects on the product being developed, on the process to

develop it, and on the organization as a whole. Indeed, the price for it is the mentioned requirement for upfront investments, which can be recognized both in term of direct and indirect perceptions. It can be perceived directly in terms of more resources, extra complexity of the structure, and a sense of decreased productivity. It can be perceived indirectly in terms of the *feelings* of lack of autonomy, of the top-down planning, and of the lack of flexibility, which are direct dependant from the global organization, which naturally make the decision making process less decentralized but, if done properly, more cohesive. Here we emphasize the term *feelings* because it is our *impression* (also emphasized) that these are structural issues of agile teams, which can be handled with more investments in creating an overall corporate culture, as it now becoming evident in other industry sectors [24, 61, 62, 66].

We can summarise of findings coming from the multivocal literature review and the empirical investigation (Table 8) suggesting that:

- **SAFe requires an upfront commitment** of the organization as a whole, which also imply the effort of educating all the workforce and the ability and possibility to wait for it to take effect.
- **SAFe is agile** in the traditional sense of the word, that is the individual practices of SAFe do not appear to hinder agility overall, even if SAFe has requirements to succeed that seem to conflict with the vulgata on agility.
- **SAFe carries benefits** for the organizations adopting it in terms of **productivity**, improved **cohesion** of the company, more **effective organization**, and **quality**, and to **raises challenges** for the **extra resources** needed and for the requirement to properly educate the whole workforce and establishing a suitable company culture; however, such kind of extra resources appear now to be an emerging requirement for companies moving to lean development processes.

7 Threats to validity

Indeed, all works based on empirical investigations are subject to risk of invalid conclusions, and this is not an exception to the norm.

Our work has two components: a multivocal literature review and an empirical study. We have explained above in Section 2 how we have performed the MLR, hoping to adhere to the highest scientific standards.

As far as the experimental part is concerned, our work is mainly observational, ruling out the possibility of performing inference, therefore, the discussion on validity becomes mostly qualitative. In terms of internal validity, the questionnaire and the interviewing processes have been organized to minimize the risk of researchers and respondent biases. For the external validity and the generalizability of findings, our dataset is in our view representative but it is also not very large: we used LinkedIn to find possible professionals to interview; as usual it is difficult to find people willing to answer research questionnaires; this has been even more challenging because our focus was the small community of managers and developers using or interested in SAFe. Still, we have minimal coverage error because the participants cover all levels of SAFe; however, the non-response error was high as none from the portfolio level

Table 8: Overall summary of the answers to the research questions

Research question	Answers from the MLR	Answers from the empirical investigation
RQ ₁ : Success factors of SAFe	Company support, and motivated and competent team, supplemented by adequate coaching	Confirmed and emphasized that the first and foremost reason for success of SAFe lies in a comprehensive and informed support from the organization at all its level.
RQ ₂ : SAFe hinders agility?	The alleged complex structure of SAFe and the interpretation of alignment may hinder agility, especially as reported by practitioners in gray literature.	Rather than hindering agility, SAFe appears to expose its essence, that is, an approach that requires a substantial investment.
RQ ₃ : Effects of SAFe	<i>Benefits</i> : Overall, better business cohesion, operational structure, and quality of the product <i>Drawbacks</i> : More difficult and heavy-weighted process, with negative consequences on activities requiring changes, like planning, process improvement, decision making, communications.	Remarkable positive effects on the product being developed, on the process to develop it, and on the organization as a whole. Upfront company wide investments, both direct, e.g., more resources and indirect, e.g., apparent lack of autonomy and flexibility.

answered. With reference to the construct validity, we elaborated our questionnaire on the basis of the GQM, based in turn on the results of the MLR.

Needless to say, that this work is intended to pave the way of followup replications and experimentations, extending the number of respondents and their diversity in terms of location, company structure, individual background, and so on.

8 Conclusions and further research

This paper has discussed the issues that surround the transformation towards SAFe, one of the most notable attempts to scale agile methods to large organizations. The key research questions that have been analysed are (RQ₁) the success factors for SAFe, (RQ₂) whether SAFe hinders agility, and (RQ₃) the overall effects of the introduction of SAFe. The study is fundamentally exploratory and observational, with the intention of being a first comprehensive study on the matter for followup replication and deepening. It has been organized in a multivocal literature review that has identified a set of preliminary answers to three research questions following the example of [57], and then an empirical analysis based on questionnaires that were answered by 25 respondents from 17 different companies located in eight countries.

The results appears to confirm the validity of SAFe overall as a company-wide approach to agile development.

This paves the way for essential replications, possibly also using meta-analysis [20], for which it is important not only to just strengthen the conclusions found in this paper but also to analyse all the collateral factors that may play an essential role in providing precise answers to the mentioned research questions, such as the background of the people involved, their education, their working environment, their fundamental cultural values etc., which are all now emerging as key aspects in the introduction of any lean-based approaches, as agile (and among agile SAFe) [8, 17, 32]. To promote such replications, all the data used in paper will be made available in anonymous form to protect the privacy of the respondents.

It would also be very important to perform similar studies for other agile frameworks applicable in large organizations, such as LeSS or RAGE. In this way there will be the possibility of performing a comparative analysis of the different approaches to gather

further understanding of their strengths, their weaknesses, and their degrees of applicability.

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References

- [1] Samia Abdalhamid and Alok Mishra. 2017. Adopting of Agile Methods in Software Development Organizations: Systematic Mapping. *TEM Journal* 6, 4 (2017), 817–825.
- [2] Lyssa Adkin. 2014. *The Agile Coaches' Coach Shares Her View on SAFe*. <https://www.infoq.com/articles/agile-coaches-coach-view-safe/>, visited 2021-10-11.
- [3] Pauliina Ahlgren. 2016. *Managing a successful SAFe transformation*. Bachelor. University of applied science.
- [4] Victor R. Basili, Gianluigi Caldiera, and H. Dieter Rombach. 1994. The Goal Question Metric Approach. In *Encyclopedia of Software Engineering*. Wiley.
- [5] Daniel Beal and Jeremy Dawson. 2007. On the Use of Likert-Type Scales in Multilevel Data. *Organizational Research Methods* 10 (07 2007), 657–672. <https://doi.org/10.1177/1094428106295492>
- [6] P. C. Beatty and G. B. Willis. 2007. Research Synthesis: The Practice of Cognitive Interviewing // Research Synthesis. *Public Opinion Quarterly* 71, 2 (2007), 287–311. <http://poq.oxfordjournals.org/cgi/doi/10.1093/poq/nfm006>
- [7] P. M. Bentler, Douglas N. Jackson, and Samuel Messick. 1971. Identification of content and style: A two-dimensional interpretation of acquiescence. *Psychological Bulletin* 76, 3 (1971), 186–204. <https://doi.org/10.1037/h0031474>
- [8] Sanjay Bhasin. 2013. Impact of corporate culture on the adoption of the Lean principles. *International Journal of Lean Six Sigma* 4 (05 2013), 118–140. <https://doi.org/10.1108/20401461311319329>
- [9] Trevor G Bond and Christine M Fox. 2013. *Applying the Rasch model: Fundamental measurement in the human sciences*. Psychology Press.
- [10] Harry N. Boone and Deborah A. Boone. 2012. Analyzing Likert data. *Journal of Extension* 50, 2 (2012), 1–5.
- [11] Norman M. Bradburn and Seymour Sudman. 2004. *The Current Status of Questionnaire Research*. Wiley, Chapter 2, 27–40.
- [12] Patrik Carlson. 2017. *Scaling Agile Software Development: A Multiple Case Study of the Scaled Agile Framework*. Master's thesis. Aalto University.
- [13] Paolo Ciancarini, Mirko Farina, Sergey Masyagin, Giancarlo Succi, Sofia Yermolaieva, and Nadezhda Zagvozkina. 2021. Non Verbal Communication in Software Engineering – An Empirical Study. *IEEE Access* 9 (2021), 71942–71953. <https://doi.org/10.1109/ACCESS.2021.3075983>
- [14] Paolo Ciancarini, Sergey Masyagin, Victoria Rotaru, and Giancarlo Succi. 2020. Analysis of the working patterns during the COVID-19 pandemic - an industrial case study. In *2020 International Computer Symposium (ICS)*. 497–502.
- [15] Luis Corral, Alberto Sillitti, and Giancarlo Succi. 2013. Agile Software Development Processes for Mobile Systems: Accomplishment, Evidence and Evolution. In *Mobile Web Information Systems*, Florian Daniel, George A. Papadopoulos, and Philippe Thiran (Eds.). Springer Berlin Heidelberg, Berlin, Heidelberg, 90–106.
- [16] John W. Creswell. 2013. *Qualitative Inquiry and Research Design*. SAGE Publications, 83–104.
- [17] Kimberly Deranek, Shweta Chopra, and Gretchen Mosher. 2017. Lean adoption in a small and medium enterprise: Model validation. *Journal of Technology, Management, and Applied Engineering* 33 (09 2017), 1–13.

- [18] Digital.AI. 2021. *15th Annual State of Agile Report*. <https://stateofagile.com/>, visited 2021-10-11.
- [19] Torgeir Dingsøy, Nils Brede Moe, and Helena Holmstrom Ohlsson. 2018. Towards an understanding of scaling frameworks and business agility: a summary of the 6th international workshop at XP2018. In *Proceedings of the 19th International Conference on Agile Software Development*. ACM, 6:1–6:4.
- [20] Gcinizwe Dlamini, Firas Jolha, Zamira Kholmatova, and Giancarlo Succi. 2022. Meta-analytical comparison of energy consumed by two sorting algorithms. *Information Sciences* 582 (2022), 767–777. <https://doi.org/10.1016/j.ins.2021.09.061>
- [21] Tore Dybå and Torgeir Dingsøy. 2008. Strength of Evidence in Systematic Reviews in Software Engineering. In *Proceedings of the Second ACM-IEEE International Symposium on Empirical Software Engineering and Measurement*. Association for Computing Machinery, 178–187.
- [22] Christof Ebert and Maria Paasivaara. 2017. Scaling agile. *IEEE Software* 34, 6 (2017), 98–103.
- [23] Kyle Evans. 2019. *The Major Problems with SAgE*. <https://productcoalition.com/the-major-problems-with-safe-1e797f7e48f8>, visited 2021-10-11.
- [24] Mirko Farina, Marina Ivanova, Maxim Korsunov, Alexander Krivosov, Artem Kruglov, Nataliya Matrosova, and Giancarlo Succi. 2021. Musical Practices in Software Development: Insights from Gary Marcus's Guitar Zero. In *2021 International Conference "Nonlinearity, Information and Robotics" (NIR)*. 1–7. <https://doi.org/10.1109/NIR52917.2021.9666082>
- [25] Martin Fowler. 2004. C3. <https://www.martinfowler.com/bliki/C3.html>, visited 2021-10-11.
- [26] Sandra Fridålv. 2017. *Evaluating the implementation of SAgE with focus on test and quality*. Master's thesis. Lund University.
- [27] Adrian Furnham. 1986. Response bias, social desirability and dissimulation. *Personality and individual differences* 7, 3 (1986), 385–400.
- [28] Vahid Garousi, Michael Felderer, and Mika V. Mäntylä. 2019. Guidelines for including grey literature and conducting multivocal literature reviews in software engineering. *Information and Software Technology* 106 (2019), 101–121.
- [29] Vahid Garousi, Michael Felderer, and Mika V. Mäntylä. 2019. Guidelines for including grey literature and conducting multivocal literature reviews in software engineering. *Information and Software Technology* 106 (2019), 13.
- [30] Daniel Gerster and Christian Dremel. 2018. Agile Meets Non-Agile: Implications of Adopting Agile Practices at Enterprises. In *24th Americas Conference on Information Systems*. Association for Information Systems, 1–10.
- [31] Paul Hands. 2018. The Scaled Agile Framework – When it works, it works well. When it doesn't work, it's waterfall. <https://blog.scottlogic.com/2018/05/14/the-scaled-agile-framework.html>, visited 2021-10-11.
- [32] Ulrika Harlin, Promporn Wangwacharakul, Martina Berglund, and Per Gullander. 2014. Cultural Aspects when Implementing Lean Production and Lean Product Development - Experiences from a Swedish Perspective. *Quality Innovation Prosperity* 18 (06 2014), 321. <https://doi.org/10.12776/QIP.V18I1.321>
- [33] Saara Hynninen. 2019. *Benefits of Scaled Agile Framework (SAgE)*. <https://www.tieto.com/en/blog/2019/06/benefits-of-scaled-agile-framework-safe/>, visited 2021-10-11.
- [34] Vladimir Ivanov, Alan Rogers, Giancarlo Succi, Jooyong Yi, and Vasili Zorin. 2017. What Do Software Engineers Care About? Gaps Between Research and Practice. In *Proceedings of the 2017 11th Joint Meeting on Foundations of Software Engineering (Paderborn, Germany) (ESEC/FSE 2017)*. ACM, New York, NY, USA, 890–895. <https://doi.org/10.1145/3106237.3117778>
- [35] Andrea Janes, Alberto Sillitti, and Giancarlo Succi. 2013. Effective dashboard design. *Cutter IT Journal* 26, 1 (2013), 17–24.
- [36] Andrea Janes and Giancarlo Succi. 2014. *Lean Software Development in Action*. Springer, Heidelberg, Germany. <https://doi.org/10.1007/978-3-642-00503-9>
- [37] Ron Jeffries. 2014. *Issues with SAgE*. <https://ronjeffries.com/xpro/articles/issues-with-safe/>, visited 2021-10-11.
- [38] Ron Jeffries. 2014. *SAgE - Good But Not Good Enough*. <https://ronjeffries.com/xpro/articles/safe-good-but-not-good-enough/>, visited 2021-10-11.
- [39] Martin Kalenda, Petr Hyna, and Bruno Rossi. 2018. Scaling agile in large organizations: Practices, challenges, and success factors. *Journal of Software: Evolution and Process* 30, 10 (2018), e1954.
- [40] B. Kitchenham and S. Charters. 2007. *Guidelines for performing Systematic Literature Reviews in Software Engineering*. Technical Report. EBSE 2007-001. Keele University and Durham University Joint Report. 1–44 pages.
- [41] Artem Kruglov. 2021. Impact of the Communication Issues: A Case Study of IT Start-Up. In *Frontiers in Software Engineering*. Springer International Publishing, 113–132.
- [42] Artem Kruglov, Dragos Strugar, and Giancarlo Succi. 2021. Tailored performance dashboards—an evaluation of the state of the art. *PeerJ* 7 (Oct. 2021), e625.
- [43] Maarit Laanti and Petri Kettunen. 2019. SAgE Adoptions in Finland: A Survey Research. In *Agile Processes in Software Engineering and Extreme Programming – Workshops*. Springer, 81–87.
- [44] Marc Lankhorst. 2019. Agile Architecture: Using ArchiMate with the Scaled Agile Framework (SAgE). <https://bizdesign.com/blog/agile-architecture-using-archimate-with-the-scaled-agile-framework-safe/>, visited 2021-10-11.
- [45] Paul S. Levy. 2005. Cluster Sampling. In *Encyclopedia of Biostatistics*. American Cancer Society. <https://doi.org/10.1002/0470011815.b2a16012> arXiv:<https://onlinelibrary.wiley.com/doi/pdf/10.1002/0470011815.b2a16012>
- [46] Peter Lynn and Olena Kaminska. 2012. Factors Affecting Measurement Error in Mobile Phone Interviews. In *Telephone Surveys in Europe: Research and Practice*, Sabine Häder, Michael Häder, and Mike Kühne (Eds.). Springer, 211–228.
- [47] Microsoft. [n.d.]. *LinkedIn*. [linkedin.com](https://www.linkedin.com)
- [48] Model systems knowledge transaction center. 2015. A Guide for Developing a Protocol for Conducting Literature Reviews . , 2 pages.
- [49] Leo Nyholm. 2018. *SAgE model and using it in different circumstances*. Master's thesis. University of Vaasa.
- [50] Maria Paasivaara. 2017. Adopting SAgE to Scale Agile in a Globally Distributed Organization. In *2017 IEEE 12th International Conference on Global Software Engineering (ICGSE)*. IEEE, 36–40.
- [51] Aditya Pancholi and Sapna Grover. 2014. Scaled Agile Framework: A Blight. In *International journal of innovative research and development*, Vol. 3. Globeedu Group, 425–427. Issue 5.
- [52] M.Q. Patton and Ralph Erskine Conrad Memorial Fund. 2002. *Qualitative Research & Evaluation Methods*. SAGE Publications, 209–259.
- [53] A Pitkänen. 2015. *Agile Transformation: A case study*. Master's thesis. Aalto university.
- [54] Philip M. Podsakoff, Scott B. MacKenzie, Jeong Yeon Lee, and Nathan P. Podsakoff. 2003. Common Method Biases in Behavioral Research: A Critical Review of the Literature and Recommended Remedies. *Journal of Applied Psychology* 88, 5 (10 2003), 879–903.
- [55] Jan Pries-Heje and Malene M. Krohn. 2017. The SAgE Way to the Agile Organization. In *Proceedings of the XP2017 Scientific Workshops*. Association for Computing Machinery, 18:1–18:3.
- [56] Abheeshta Putta, Maria Paasivaara, and Casper Lassenius. 2018. Adopting scaled agile framework (SAgE). In *Proc. 19th Int. Conf. on Agile Software Development Companion - XP '18*. ACM Press, 1–4.
- [57] Abheeshta Putta, Maria Paasivaara, and Casper Lassenius. 2018. Benefits and Challenges of Adopting the Scaled Agile Framework (SAgE): Preliminary Results from a Multivocal Literature Review. In *Product-Focused Software Process Improvement*, Marco Kuhrmann et al. (Eds.). Springer, 334–351.
- [58] Scaled Agile. 2018. Scaled Agile Framework - Safe for Lean Enterprises . <https://www.scaledagile.com/resources/safe-whitepaper/>, visited 2021-10-11.
- [59] Philip Sedgwick. 2013. Stratified cluster sampling. *BMJ* 347 (2013). <https://doi.org/10.1136/bmj.f7016> arXiv:<https://www.bmj.com/content/347/bmj.f7016.full.pdf>
- [60] M. Shameem, C. Kumar, B. Chandra, and A. A. Khan. 2017. Systematic Review of Success Factors for Scaling Agile Methods in Global Software Development Environment: A Client-Vendor Perspective. In *24th Asia-Pacific Software Engineering Conference Workshops*. IEEE, 17–24.
- [61] Mehra Taherimashhadi and Imma Ribas. 2018. A Model to align the organizational culture to Lean. *Journal of Industrial Engineering and Management* 11, 2 (2018), 207–221.
- [62] Rio Tanudiharjo, Florence Yun, John Joo, and Ivan Arokiam. 2021. Investigation of Factors Impacting Lean Implementation in the Indonesian Fast-Moving Consumer Goods Industry. *Operations and Supply Chain Management: An International Journal* 14, 2 (2021), 162–172. <https://doi.org/10.31387/oscm0450294>
- [63] David L. Vannette and Jon A. Krosnick. 2014. *Answering Questions: A Comparison of Survey Satisficing and Mindlessness*. John Wiley & Sons, Ltd, 312–327.
- [64] Sina Katharina Weiss and Philipp Brune. 2017. Crossing the Boundaries – Agile Methods in Large-Scale, Plan-Driven Organizations: A Case Study from the Financial Services Industry. In *Proc. 29th Int. Conf. on Advanced Information Systems Engineering*. Springer, 380–393.
- [65] Virginia Wilson. 2014. Research Methods: Triangulation. *Evidence Based Library and Information Practice* 9 (03 2014), 74–75. <https://doi.org/10.18438/B8WW3X>
- [66] Meiling Wong. 2007. The Role of Culture in Implementing Lean Production System. In *Advances in Production Management Systems*, Jan Olhager and Fredrik Persson (Eds.). Springer US, Boston, MA, 413–422.
- [67] R.K. Yin. 2012. *Applications of Case Study Research*. SAGE Publications, 8–15.